Chapter 27

Adult Learning Theories in Pharmacy Education

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Learning objectives

Objective 27.1 Outline the concepts in adult learning relevant to pharmacy education.
Objective 27.2 Explain applications of adult learning principles in pharmacy education.
Objective 27.3 Describe how to write learning objectives in pharmacy education.

OBJECTIVE 27.1. OUTLINE THE CONCEPTS IN ADULT LEARNING RELEVANT TO PHARMACY EDUCATION

Learning involves acquiring knowledge, skills, and competence. Knowledge is an essential component in deciding, performing, and reviewing actions. Actions are properly implemented with skills (psychomotor skills). Skills are also involved in how knowledge is used (cognitive skills). Values and attitudes contribute to reasoned action and competency to practice. To practice the profession of pharmacy, a student should learn stipulated knowledge, skill, values, and attitudes. Learning starts with receiving or finding information. However, information does not become learning until an individual processes it and makes it his or her own. Additionally, learned information must be used regularly for knowledge retention to occur. Knowledge can also be refreshed and updated as needed. As students progress from being novice to expert, they are able to learn independently because they have acquired the cognitive knowledge and are proficient in the relevant physical, social, and psychological environment of professional practice. However, acquiring professional expertise can take thousands of hours, guided practice, and the willingness to learn from experience.[1]

Adult learning theories evolved over time, with theories specifically explaining how adults learn emerging in the 1970s. Traditional learning theories tend to be individualistic. Examples include a concept of learning as a change in behavior (behaviorism), as personal development (humanism), as a mental process (cognitivism), and as creating meaning from experience (constructivism).

- Behaviorism: Behaviorism focuses on observable behaviors that are shaped by the environment, which is designed by the teacher. Learning is controlled by the positive and negative reinforcements the teacher places in the environment. Behaviors are broken down into smaller tasks, with a focus on stepwise mastery.[2] Behaviorism is operating in pharmacy skills labs: students master taking blood pressure by practicing each required subtask until they develop mastery. Often, these types of skills are evaluated with rubrics in which addition or subtraction of points is the reinforcing locus of control. Sometimes, these are even high stakes, pass or fail behaviors.
- Humanism: The theory of humanism posits that learning occurs to fulfill an individual's potential, to achieve self-actualization. As such, learners are motivated from within. The learning environment must be learner-centered, with educators facilitating students' self-discovery and creativity.[3] Self-directed learning evolved from humanism and is essential to incorporate into pharmacy curricula, so graduates can begin the continual professional development required throughout their careers. An example includes self-discovery and portfolio courses, in which students identify their own learning needs, select their own learning activities to achieve self-written objectives, and document and

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reflect on the outcomes. Andragogy, the education of adults, also has roots in humanism. Andragogy holds that adult learners are independent and self-directed, are intrinsically motivated to learn, are practical and problem-focused, have rich experiences to learn from, and value learning when it is congruent with their daily life.[4] Beyond andragogy and heutagogy, the study of self-determined learning seeks to create learners highly capable and competent of autonomous self-directed learning.[5]

- Cognitivism: While behaviorism focuses on learning shaped by the external environment, and what behaviors can be observed, cognitivism focuses on the internal environment of the learner's mind, thought, and memory. Cognitivism requires educators and learners to focus on how dynamic memory and thought processes are effectively changed. As such, reflection and retrieval practice are helpful tools to achieve learning outcomes in the cognitivist framework.[6] Educators must be aware of cognitive load and threats to building long-term memory and critical thinking skills. Metacognition, the awareness of and ability to manage one's own learning and thinking processes, is viewed as essential to successful learning in cognitivism and can be supported through instructional strategies and modeling.[2,7] In pharmacy education, cognitivism can be seen in much of the curriculum, from teaching medication facts to teaching how to reconcile patient-specific factors, including patient preferences, drug-specific factors, and evidence to select a safe and effective medication for a given patient.

Cognitive learning requires thoughtful, focused efforts to relate new and old knowledge and experience. Concept maps help activate higher-order neural networks. Challenging students to create a concept map of their knowledge can help them integrate and reconcile new information and constructs with prior learning to build long-term memory of complex information. Concept maps can mitigate students' tendency to employ rote-learning strategies. This active learning strategy requires students to graphically represent concepts and the interrelationships among them. Instructors can scan a student's concept map to quickly identify and correct any misconceptions held by students.[8]

- Constructivism: Constructivist theory holds that knowledge and understanding is attained through experience and action.[9] It relates to how individuals integrate new knowledge, reconcile it with existing knowledge, and change or expand their mental schemas, meaning "knowledge structures that represent repeated patterns of experiences (cognitive as well as behavioral)".[1] Teaching methods that build on identifying what students already know can help them establish cognitive connections between what they know and what is to be learned. Constructivism is learner-centered and focused on discovery. For example, assign students a drug information question on a clinically controversial topic, followed by a reflective writing prompt and group discussion. This activity would foster students' discovery and reconciliation of contradictions between new information and their previous knowledge and experience on the topic.

Recent adult learning theories are more representative of people living and working together within a sociocultural and historical context. These include situated cognition, social cognitive learning, and communities of practice.

- Situated cognition: Schumacher et al.[6] contend that situated cognition is the ideal theory of learning in medical fields. Situated cognition grew out of constructivism. While constructivism focuses on the individual learner constructing his or her own knowledge, situated cognition maintains that the individual's construction of knowledge hinges on social and cultural interactions of the learner with his or her environment. Learning and knowledge are inseparable from the context, activity, and tools used in that activity; people pharmacists interact within their work, culture, and language. Learning is shaped by social context and the tools used in that setting; learning is seen as an increasingly effective performance across time and situations. A related concept is *scaffolding of learning*, in which pharmacy students are provided support including relevant examples, guides, templates, and other resources. As the student becomes more proficient at the assigned task, the supports are gradually removed. Skills lab makes use of this idea as students learn to handle prescriptions, drugs, bottles, labels, and counting trays. Students receive guidance as they learn to practice each skill independently, but over time they are expected to integrate related skills, become more agile in their performance, and the guidance is gradually reduced. Not surprisingly, pharmacy students who have served as pharmacy technicians have an advantage at first.
- Social Cognitive Learning and Communities of Practice: Communities of practice evolved from social learning theory, the idea that learning is social versus an individual endeavor. In a community of practice, people engage in a network of learning with others in a shared domain of competence or endeavor.[10,11] Newcomers learn from others, but over time they become contributing members and more "senior" members of the group who may in turn learn from new members. There are many online communities of practice in pharmacy education, including the formal communities of practice, such as the American College of Clinical Pharmacy's Practice and Research Networks, and informal, organic communities of practice that emerge on social networks such as Twitter. A related concept is *cognitive apprenticeships* in which apprentices are taught to think about what they are doing as they learn to perform actions; the apprentice serves

under a master who can model appropriate thinking/behaviors/actions and teach what the apprentice should learn and be able to perform. This is related to the clerkship and student/preceptor relationship in which the student learns to perform professional responsibilities under the supervision of a licensed pharmacist.

Practice exercise—structure a group learning activity that requires students to depend on one another:

Create a problem-based learning (PBL) scenario that involves students in solving a group challenge in which the initial information provided is ill-defined. Employ information from authentic practice situations. Students are expected to use questioning, information retrieval strategies, clinical knowledge, and reasoning to make recommendations. PBL requires information analysis and synthesis as well as iterative discussion. Instructors guide the reasoning and deliberative processes involved in students' development of a detailed, stepwise proposal for addressing a patient problem. Instructors challenge the depth and accuracy of their approach, and avoid solving it for them. PBL promotes students' self-directed learning capabilities and part of the instructors' goal is to promote that ability.[12,13] These very skills are needed during training in which students learn they cannot rely on a preceptor's knowledge alone but develop their own methods to figure out the situation. Students must use skills of information retrieval and clinical reasoning to suggest their pharmacy recommendations for patient cases to preceptors.

OBJECTIVE 27.2. EXPLAIN APPLICATIONS OF ADULT LEARNING PRINCIPLES IN PHARMACY EDUCATION

Andragogy, meaning the education of adults, holds that adults learn differently from children (Table 27.1). Malcom Knowles, the father of andragogy, proposed that the defining characteristics of an adult learner are psychological (self-direction, personal responsibility). Adult educators tend not to define adulthood chronologically, but rather as occurring when individuals take on the responsibilities of adult life, e.g., holding a responsible job, supporting oneself, finding a spouse, having children.[4] Many pharmacy students may not meet this definition of adulthood at matriculation. As a concept, andragogy can be used to help pharmacy instructors promote self-direction and a responsible adult self-concept among students through curriculum and instructional design. In addition, pharmacy students should learn about adragogy to become an effective patient educator. To connect well with an adult learner, the pharmacy information being conveyed should be engaging, personally relevant, and immediately useful.[4]

In pharmacy education, curricula must be mapped to outcomes outlined by accrediting bodies to maintain quality standards across educational institutions, preventing learning from being completely student-driven. However, it is possible to incorporate andragogy into instructional design of pharmacy curricula by incorporating the principles outlined in Table 27.2. Students often exhibit more self-directed learning capacity in the informal or hidden curriculum in pharmacy school by choosing which professional organizations to work within and which elective curricular and extracurricular educational activities they will engage in. Self-directed learning could also be encouraged by allowing students to choose among learning activities that support the same learning objectives, such as providing a choice between writing a scholarly paper or preparing a presentation. Examples of smaller assignments include providing a choice between reading a chapter versus watching YouTube videos to prepare for class.

In addition to Knowles, Paulo Freire and Jack Mezirow further shaped adult learning theory. Freire believed education is a tool for social change and focused on shifting power from teachers to learners. In *Pedagogy of the Oppressed*, he proposed a collaborative, horizontal relationship between educators and learners, in which learners contribute to and direct the learning environment. Mezirow advocated that educators have a responsibility to facilitate development of

TABLE 27.1 Assumptions of Pedagogy and Andragogy[4]

Assumption	Pedagogy	Andragogy
Need to know	Teacher/grade-centered	Learner-centered/relevance
Learner's self-concept	Dependence (on educator)	Autonomy and personal responsibility
Role of experience	Educator's wisdom through experience is paramount	Integrating learners' experience is paramount
Readiness to learn	Ready to learn what teacher thinks is important, required knowledge to pass or get desired grade	Ready to learn what is relevant to life and role
Orientation to learning	Subject-oriented	Task- and problem-oriented
Motivation	External (i.e., grades) > internal	Internal > external

TABLE 27.2 Implications of Adult Learning Principles in Pharmacy Education[4]

Principle of Adult Learning[9]	Implication for Pharmacy Education
Adults need to understand why they should learn something.	If pharmacy students fail to see the relevance of content to modern pharmacy practice, they will disengage. For example, modern pharmacy curricula should not include content on medications that are no longer marketed unless the instructor is specifically and explicitly relating that content to a modern pharmacy problem, such as a parallel or recurrent theme in pharmacy practice. Short cases, simulations, and storytelling (pharmacy history, pharmacy law, medical humanities) may improve connection to a topic.
Adults have an adult self-concept and are self-directed individuals.	Adult learners resent situations in which they feel their autonomy is challenged. However, pharmacy students may default to the pedagogical hallmark of passivity in learning environments due to habit. Some students also will not be fully developed self-directed learners at matriculation. Educators should design instruction for pharmacy students that fosters self-directed learning. Creating "flipped classrooms" and incorporating active learning strategies supports self-concept and self-directed learning.
Adult learners are not blank slates; they have knowledge and life experience.	Provide opportunities for pharmacy students to share, apply, or build on their experiences, whether they are work-related (e.g., pharmacy technician or research) or personal experiences with illnesses relevant to course content.
Adult learners are self-directed related to the developmental tasks of his/her social role.	The developmental tasks of a pharmacy technician, student, and pharmacist are vastly different. To overcome the tendency for adults to prefer learning knowledge and skills applicable to their current role and, daily life, pharmacy educators may need to use simulation, shadowing, and career interviews as instructional strategies.
Adult learners are looking for immediate application; learning is relevancy-oriented.	Students will remember the information they believe is significant. Pharmacy educators should consider using pharmacist testimonials, patient testimonials, and simulation to ensure pharmacy students understand the relevance of content. Curriculum and instructional designers should consider the timing of the content. For example, focusing on medication history-taking just prior to an introductory rotation where a student can perform medication histories would be consistent with this principle.
Adults tend to be driven by internal motivation.	Be aware that assessments that foster competition for external rewards may undermine the ability of pharmacy students to remain internally motivated. Ask students to set personal goals and intentions unrelated to external motivators at the beginning of a course or unit. (e.g., evaluate my dad's inhaler technique and provide relevant inhaler education to my dad vs. Make an A on the asthma exam.) Consider using formative, nongraded assessments and reflections to foster internal motivation and focus on personal growth.

learners' critical thinking skills. That is, learners should critically reflect on their own assumptions, including the consequences of those assumptions, and reconcile the validity of held assumptions against alternative assumptions.[14] Discussion, reflective writing, debate, and simulation with standardized patients can be used in the modern pharmacy classroom to facilitate development of critical thinking skills. In an activity based on the Think, Pair, Share strategy, students reflect, write, and discuss their ideas with a partner before sharing them with the larger group. Both Freire and Mezirow were proponents of critical reflection, but Freire thought reflection should be balanced with acting on the world. In modern pharmacy education, those educators embracing learner-centered teaching are a "guide on the side," facilitating student-driven learning, in the spirit of Freire and Mezirow, versus a "sage on the stage," where the professor primarily transfers information to students through lecturing.[14,15]

Pharmacy education should start with the end in mind.[16] Determine which theories and teaching methods will most support pharmacy students achievement of student learning outcomes. The learning objectives are planned to achieve the corresponding learning outcomes. The objectives are mainly planned by the teachers in discussion with students and the outcomes are achieved by the students. In problem-based learning students develop learning objectives. Achievements of learning outcomes to be confirmed with appropriate assessment tools. Sometimes learning objectives and learning outcomes are used interchangeably. Consider characteristics of the audience, such as age/generation, gender, and socioeconomic backgrounds. Although popular in the literature, tailoring instruction to specific learning styles has not been shown to improve learning.[17] Regardless of the learning theory and related instructional methods selected, consider that

accumulating evidence shows that learner-centered methods such as retrieval practice, reflection, and flipped classrooms positively impact pharmacy student engagement and learning.[18–20]

Practice exercise—A persistent challenge in educating pharmacy students is retention of information for later application in practice. Imagine a senior pharmacy student in one of his/her final clerkship assignments who is reluctant to search out information about relevant medications relevant to patient cases he/she is responsible for. Instead, he or she appears too dependent on the clinical preceptor for medication information. Using the information in this chapter, think about ways you could help him or her move to become self-directed in his/her learning so that he/she can transition to the next phase of his/her career, in which he/she will be a licensed pharmacist and need to work more independently.

OBJECTIVE 27.3. DESCRIBE HOW TO WRITE LEARNING OBJECTIVES IN PHARMACY EDUCATION

Adult learning suggests that learning must be relevancy-oriented. Learning objectives communicate the relevancy of content to pharmacy students and educators. Learning objectives in pharmacy education are layered. Terminal course-level learning objectives should map to the terminal performance outcomes of the pharmacy curriculum. Likewise, learning objectives written by individual educators for single course sessions or units of instruction within a course should support achievement of, and map to, course-level terminal learning outcomes.[16] To avoid cognitive overload, educators should focus on three to five learning objectives per learning activity. However, students may need to write their own supporting, or enabling learning objectives to ensure they gain the requisite knowledge, skills, or attitudes to meet a learning objective written by an instructor.[16] Students must also write their own learning objectives in self-directed learning courses, such as some professional development, advanced practice pharmacy experience, and portfolio courses. Effective learning is built on a foundation of quality learning objectives aligned with teaching and learning methods as well as assessments.

Use the simple model A-B-C-D to write learning objectives, but also make them SMART (specific, measurable, achievable, relevant, and time-bound).[16] An example objective that encompasses these points would be: "Given an e-learning module, and case discussion, the PHAR 501 student should be able to list five patient-related variables that increase the risk of adverse effects from opioid analgesics." Table 27.3 outlines the construction of objectives.

Desired learning outcomes often encompass multiple domains or dimensions of learning, including cognitive, affective, psychomotor, and interpersonal.[16] Consider the following scenario: You are assigned to teach blood pressure measurement to pharmacy students. As you think through what they should be able to do at the end of your session, you want them to be able to measure a blood pressure, record it, and explain a patient's blood pressure reading to that patient. Measuring blood pressure requires a variety of skills including selecting an appropriate cuff size, inflating and deflating the cuff, and measuring the pulse. These skills are all in the psychomotor dimension of learning, which are manual or physical skills.

TABLE 27.3 Writing Effective Learning Objectives[16]

		Good Examples	Bad Examples
A	Who is your Audience? Be **specific** and write for a singular learner.	• PHAR 501 student • City Pharmacy intern • Midyear attendee	• Students • Learners • Participants
B	What Behavior is desired? It must be observable, **relevant, measurable**, and **achievable** given the resources of your learning activity (**time**, type, etc.).	• Discuss • Identify • Formulate	• Understand • Learn • Know
C	Under what Conditions should the audience achieve this behavior?	• Given a reading, discussion, and role play … • Given access to tertiary databases … • Given a simulated patient with hypertension …	• On completion of this session …
D	To what Degree (extent, measure, or scope) should the behavior be performed? This is an evaluative statement. Assessments should reflect this statement. This statement is often omitted in practice if achievement of an objective will not be evaluated.	• By offering an opinion at least twice • Within 15 min • Without error	• Successfully • Safely • Carefully • At discretion of the examiner • Until finished

Students will also need to have knowledge of what blood pressure is, how to use the cuff and stethoscope, how to interpret systolic and diastolic readings, and some knowledge about communication techniques. These are all in the cognitive dimension. Students will also need to understand the importance of pharmacist-driven blood pressure management, which is an attitude, in the affective dimension. In addition to these dimensions of learning, students will need to respond to patient cues and tailor their interaction to patient needs such as language or literacy barriers. This is the interpersonal dimension and sometimes can encompass knowledge, skills, and attitudes. Objectives should be written and assessed in a single dimension: cognitive, psychomotor, affective, or interpersonal.[16]

Benjamin Bloom created a taxonomy of learning in the cognitive dimension to systematically create learning objectives. The original taxonomy consisted of six cognitive domains. In 2001, these were modified and four knowledge domains were added.[21,22] The knowledge domains include:

- Facts: basic elements one must know to communicate with other pharmacists and solve problems in pharmacy. Examples include facts about medications, medical terminology, pharmacy laws.
- Conceptual knowledge: interrelationships between facts in a larger contextual structure. Examples include knowledge of drug classes and categories, theories, models, structures, principles, and generalizations.
- Procedural knowledge: encompasses protocols, algorithms, techniques, and methods of inquiry and procedures. Examples include how to make an IV, how to dispense a medication in a pharmacy, how to approach a drug information question, how to conduct an opioid conversion, how to interview a patient.
- Metacognitive knowledge: Self-awareness and knowledge of one's own cognition as well as knowledge of cognition in general. Examples include strategic knowledge, and awareness of how one approaches cognitive tasks and problems.[22]

Revised Bloom's Taxonomy

An example of the two-dimensional revision of Bloom's Taxonomy:

Topic: Therapeutic Use of Opioids.

Find each of the following sample behavioral statements in the table below to explore how objectives encompass the knowledge and cognitive domains of the revised Bloom's Taxonomy (Tables 27.4 – 27.7).

1. List patient-related variables that increase the risk of adverse effects from opioid analgesics.
2. Compare the efficacy of opioid analgesics for a given pain complaint.
3. Recommend an opioid analgesic based on patient- and agent-related variables.
4. Determine if opioid conversion is indicated.
5. Outline the opioid conversion process.
6. Propose an opioid stewardship policy for an ambulatory family medicine practice.
7. Interview a patient about his or her pain to collect all elements of symptom analysis.
8. Dissect your own biases in processing opioid prescriptions.
9. Select study strategies for retaining knowledge of opioid pharmacokinetics and pharmacodynamics.

Although Bloom's taxonomy for cognitive learning is the most widely known learning taxonomy, pharmacy educators and students should also be aware of Krathwohl's taxonomy for affective learning and Simpson's taxonomy for psychomotor learning. Affective learning can be the most difficult to incorporate and assess because it encompasses soft skills, but it is in the affective domain that student pharmacists develop a professional identity, empathy, and values needed for interpersonal skills. Affective learning should be integrated across pharmacy curricula.

Teaching and learning methods should align with and support learner achievement of learning objectives. Likewise, formative and summative assessments should evaluate the level and learning domain of a learning outcome.

TABLE 27.4 Blooms Taxonomy for the Opioid Example[22,23]

The Knowledge Dimension[18]	Remember	Understand	Apply	Analyze	Evaluate	Create
Factual knowledge	1					
Conceptual knowledge		2			4	3
Procedural knowledge		5	7			6
Metacognitive knowledge		9		8		

TABLE 27.5 Revised Bloom's Taxonomy for Cognitive Learning: Cognitive Domains[23]

Cognitive Domain Level	Example Action Verbs	Example Supportive Learning Activities
Remember: Access relevant knowledge (facts) from memory	Define, choose, label, list, select, identify, match	Lecture, guided note-taking, online quiz games, flashcards
Understand: Formulate meaning from factual knowledge	Compare, compute, contrast, explain, illustrate, outline, rephrase, summarize, interpret, relate, demonstrate	Guided note-taking, mindmapping or concept mapping, writing summaries, short answer questions, think-pair-share, jigsaw, reflections, infographics
Apply: Use knowledge or skill in a given situation	Apply, construct, develop, identify, interview, manipulate, plan, solve, perform	Send-a-problem, role play, fishbowls, branched e-learning scenarios, cases, simulations, skills lab exercises, formative OSCE
Analyze: Break knowledge into elemental parts and identify how they relate to each other and integrate into a larger structure	Analyze, categorize, classify, compare, contrast, diagram, discover, dissect, examine, distinguish, select	Mindmap or concept map, infographics, branched e-learning scenarios, cases, SOAP notes
Evaluate: Make and defend judgments based on standards or criteria	Assess, choose, contrast, evaluate, interpret, judge, critique, debate, measure, relate, recommend, justify, prioritize, determine, defend	Send-a-problem, debates, branched e-learning scenarios, cases, simulations, drug information exercises, formative OSCE, SOAP notes
Create: Combine elements in a different way to make something original or novel as a logical whole	Construct, create, design, develop, formulate, generate, invent, modify, plan, propose, test, solve, revise, write	Scholarly writing, research projects, cases, simulations, SOAP notes, presentations

OSCE, objective structured clinical examination; SOAP, subjective objective assessment and plan.

TABLE 27.6 Krathwohl's Taxonomy for Affective Learning[24]

Affective Domain Level	Example Action Verbs	Example Supportive Learning Activities
Receive: Becoming aware of a phenomena or situation, being open to different ideas with suspended judgment giving attention to differing ideas	Accept, notice, observe, discern, consider, listen, appreciate, realize	Patient testimonials, readings in medical humanities, activities with simulated patients, debates, role-plays as other disciplines
Respond: Voluntarily responds to a stimulus. At the highest level of responsiveness, this will be self-motivated for self-fulfillment	Engage, participate, discuss, respond, cooperate, practice, contribute	Group projects, simulations, group discussions, experiential rotations, portfolios
Value: Displays an attitude or behavior with sufficient consistency to be perceived as holding it as a value	Recognize, demonstrate, relate	Simulations, discussions, presentations, portfolios
Organize: Placing values in a system or framework and determining relationships among them	Discern, accept, reject, reconcile	Reflective writing, simulations, experiential rotations, portfolios
Internalize: Values are integrated into the person as a whole, evident in character and personality to the degree where behavior can be predicted	Display, perform	Reflective writing, simulations, experiential rotations, portfolios

Learning objectives must therefore be realistic, measurable, and achievable given curricular resources for learning activities and assessments. Although good instructional design starts with the end in mind, including identification of the minimum competencies required for a new pharmacy graduate, objectives may need modification to reflect curricular assets. For example, in a didactic course, it may be unrealistic to have this behavioral statement in a lecture-level objective: Interview a

TABLE 27.7 Simpson's Taxonomy for Psychomotor or Kinesthetic Learning[25]

Psychomotor Domain Level	Example Action Verbs	Example Supportive Learning Activities
Perceiving: Using sensory inputs to guide a motor response	Detect, distinguish, identify	Reading, watching a video, simulation
Set (mind-set): Mental, physical, and emotional readiness to act	Display, begin, proceed	Reading, watching a video (preparing)
Guided response: Imitation, trial and error, practice	Follow, reproduce, copy, replicate	Modeling following by practice in simulations, following a protocol
Mechanism (basic proficiency): Movement with some proficiency, becoming habitual	Assemble, measure, mix, display	Simulations, labs
Complex overt response (mastery): quick, accurate, automatic performance	Assemble, measure, mix, display (increasing mastery)	Simulations
Adaptation: Modification of skills for new situations or changing conditions	Adapt, alter, change, modify, revise, rearrange	Simulations
Origination: Creating new movements or new sequences of movements to address a problem	Arrange, combine, construct, create, design	Simulations

patient about his or her pain to collect all elements of symptom analysis. However, this would be a perfectly reasonable behavioral statement in an objective for an experiential rotation.

Effective pharmacy education centers on well-designed learning objectives that align with assessments and desired outcomes. Learning objectives center both educators and learners on the most relevant knowledge, skills, and attitudes to achieve desired competencies. To meet the needs of patients and healthcare teams in the 21st century, it is essential to design objectives that will support learning in the cognitive, affective, psychomotor, and interpersonal domains.

CONCLUSION

Increasing incorporation of adult learning theory throughout didactic and experiential learning may increase student engagement and improve the effectiveness of curricula. Learning objectives can ensure whether learners recognize the relevancy of designed learning activities. Depending on types of learning activities and cognitive, affective, or psychomotor learning; action verbs for learning objectives change